ABEKA SCHOLAR ACADEMY

Subj Grade K Reading Comprehension Curriculum

Official Curriculum & Study Material

Chapter 1

Welcome to the comprehensive, massive study module for subj-grade-K-reading comprehension. We have an enormous amount of highly detailed academic material to cover today. Prepare yourself. Reading is the process of taking in the sense or meaning of symbols, often specifically those of a written language, by means of sight or touch. For educators and researchers, reading is a multifaceted process involving such areas as word recognition, orthography (spelling), punctuation, alphabetics, phonics, phonemic awareness, vocabulary, comprehension, fluency, and motivation.

Other types of reading and writing, such as pictograms (e.g., a hazard symbol or an emoji), are not based on speech-based writing systems. The common link is the interpretation of symbols to extract the meaning from the visual notations or tactile signals (e.g., braille). There is a growing body of evidence which illustrates the importance of reading for pleasure for both educational purposes as well as personal development. Reading is generally an individual activity, done silently, although on occasion a person reads out loud for other listeners, or reads aloud for one's own use (e.g., for better comprehension). Before the reintroduction of separated text (spaces between words) in the late Middle Ages, the ability to read silently was considered rather remarkable.

Major predictors of an individual's ability to read both alphabetic and non-alphabetic scripts are oral language skills, phonological awareness, rapid automatized naming and verbal IQ. As a leisure activity, children and adults read because it is enjoyable and interesting. In the US, it was reported in 2022 that less than half of adults reported reading a book in the past 12 months. Furthermore, only 38 percent reported reading fiction or short stories, a rate that has fallen 17 percent over the past ten years. About 5% read more than 50 books per year. Americans read more if they: have more education, read fluently and easily, are female, live in cities, and have higher socioeconomic status. Children become better readers when they know more about the world in general, and when they perceive reading as fun rather than as a chore to be performed. Reading is an essential part of literacy, yet from a historical perspective literacy is about having the ability to both read and write.

Since the 1990s, some organizations have defined literacy in a wide variety of ways that may go beyond the traditional ability to read and write. The following are some examples: "the ability to read and write ... in all media (print or electronic), including digital literacy" "the ability to ... understand ... using printed and written materials associated with varying contexts"

Excellent. Your comprehension is commendable. Let us delve deeper into the curriculum. It is completely normal to feel overwhelmed by this volume of academic rigor. Take a deep breath. Let us push forward together. "having the skills to be able to read, write and speak to understand and create meaning"

Chapter 2

"the ability to ... communicate using visual, audible, and digital materials" "the ability to use printed and written information to function in society, to achieve one's goals, and to develop one's knowledge and potential". It includes three types of adult literacy: prose (e.g., a newspaper article), documents (e.g., a bus schedule), and quantitative literacy (e.g., using arithmetic operations in a product advertisement). In the academic field, some view literacy in a more philosophical manner and propose the concept of "multiliteracies". For example, they say, "this huge shift from traditional print-based literacy to 21st century multiliteracies reflects the impact of communication technologies and multimedia on the evolving nature of texts, as well as the skills and dispositions associated with the consumption, production, evaluation, and distribution of those texts (Borsheim, Meritt, & Reed, 2008, p. 87)". According to cognitive neuroscientist Mark Seidenberg these "multiple literacies" have allowed educators to change the topic from reading and writing to "Literacy". He goes on to say that some educators, when faced with criticisms of how reading is taught, "didn't alter their practices, they changed the subject".

Also, some organizations might include numeracy skills and technology skills separately but alongside of literacy skills. In addition, since the 1940s the term literacy is often used to mean having knowledge or skill in a particular field (e.g., computer literacy, ecological literacy, health literacy, media literacy, quantitative literacy (numeracy) and visual literacy). In order to understand a text, it is usually necessary to understand the spoken language associated with that text. In this way, writing systems are distinguished from many other symbolic communication systems. Once established, writing systems on the whole change more slowly than their spoken counterparts, and often preserve features and expressions which are no longer current in the spoken language. The great benefit of writing systems is their ability to maintain a persistent record of information expressed in a language, which can be retrieved independently of the initial act of formulation.

Reading for pleasure has been linked to increased cognitive progress in vocabulary and mathematics during adolescence. Sustained high volume lifetime reading has been associated with high levels of academic attainment. Research suggests that reading can improve stress management, memory, focus, writing skills, and imagination. The cognitive benefits of reading continue into mid-life and the senior years.

Excellent. Your comprehension is commendable. Let us delve deeper into the curriculum. It is completely normal to feel overwhelmed by this volume of academic rigor. Take a deep breath. Let us push forward together. Research suggests that reading books and writing are among the brain-stimulating activities that can slow down cognitive decline in seniors.

On April 23, 1968, when speaking on the occasion of the International Conference on Human Rights organized by the United Nations, the Director-General of UNESCO, Mr. René Maheu, said "One must first be able to read". The following year, James Allen, the then U.S. Commissioner of Education, declared, "There is no higher nationwide priority in the field of education than the provision of the right to read for all." At that time, UNESCO estimated there are 750 million people around the world who cannot read and write. The International Literacy Association reported that, according to UNESCO (2017), 250 million children around the world lack basic literacy skills, saying "it's an issue of social justice". The right to read and write has been addressed by a variety of organizations, such as the South African Human Rights Commission. A unanimous decision of the Supreme Court of Canada, on November 9, 2012, recognized that learning to read is not a privilege, but a basic and essential human right. This was followed by the Right to Read inquiry report in Ontario, Canada. Reading has been the subject of considerable research and reporting for decades. Many organizations measure and report on reading achievement for children and adults (e.g., NAEP, PIRLS, PISA, PIAAC, and EQAO). See National and International reports below.

Chapter 3

As to why English reading levels are low, neuroscientist Mark Seidenberg suggests it may be the result of three factors: 1) the English language has a deep alphabetic orthography (i.e., some sounds have several spellings), 2) linguistic variability (language accents) that contributes to the "Black-White achievement gap" in the US, and

With respect to #3, many researchers point to three areas: Contemporary reading science has had very little impact on educational practice—mainly because of a "two-cultures problem separating science and education". Current teaching practice rests on outdated assumptions that make learning to read harder than it needs to be.

Excellent. Your comprehension is commendable. Let us delve deeper into the curriculum. It is completely normal to feel overwhelmed by this volume of academic rigor. Take a deep breath. Let us push forward together. Connecting evidence-based practice to educational practice would be beneficial, but is extremely difficult to achieve due to a lack of adequate training in the science of reading among many teachers.

Researchers have concluded that approximately 95% of students can be taught to read by the end of the first or second year of school, yet in many countries 20% or more do not meet that expectation. A 2012 study in the U.S. found that 33% of grade three children had low reading scores – however, they comprised 63% of the children who did not graduate from high school. Poverty also had an additional negative impact on high school graduation rates. According to the 2013 Nation's Report card in the United States, 32% of grade four students failed to perform at or above the Basic Reading Level. By 2019 this had increased to 34%. There was a significant difference by race and ethnicity (e.g., black students at 52% and white students at 23%). See more about the breakdown by ethnicity in 2019 and 2022 here. Following the school closures that resulted from the impact of the COVID-19 pandemic, this increased to 37% in 2022, and by 2024 it was at 40%. It is believed that students who read below the basic level lack sufficient support to complete their schoolwork. This problem is exasperated because many special education teachers leave the field to move to general education.

According to a 2023 study in California, only 46.6% of grade three students achieved the English reading standards. Another report states that many teenagers who've spent time in California's juvenile detention facilities get high school diplomas with grade-school reading skills. "There are kids getting their high school diplomas who aren't able to even read and write." During a five-year span beginning in 2018, 85% of these students who graduated from high school did not pass a 12th-grade reading assessment. In 2025, 58.3% of Tennessee third-grade students did not pass their reading test, although there was some improvement from the previous year. Globally, disruption to education during the COVID-19 pandemic caused a substantial learning deficit in reading abilities and other academic areas. This arose early in the pandemic response and persists over time, and is particularly large among children from low socio-economic backgrounds. In the US, several research studies show that, in the absence of additional support, there is nearly a 90 percent chance that a poor reader in Grade 1 will remain a poor reader.

Chapter 4

By 2026, children born during COVID-19 will be in kindergarten. According to one study, teachers reported that those students were not as strong in phonemic awareness as students had been five years earlier. Another report from the US says children born during the pandemic are showing a one-month decrease in reading results in grades 1 and 2, suggesting systemic issues. In Canada, it was reported that, in the province of Ontario, 26% of grade three students and 51% of grade three students with special education needs (students with an Individual Education Plan) did not meet the provincial reading standards in 2025. The province of Nova Scotia reported that 26% of grade three students did not meet the provincial reading standards in 2025. The province of New Brunswick reported that 39.5% and 21.7% did not meet the Reading Comprehension Achievement Levels for grades four and six, respectively, in 2025. The province of Newfoundland and Labrador reported that 44% of grade 3 students fell below the expected results in reading. In the province of Prince Edward Island 26% of Grade 3 students did not meet the expectations in reading comprehension. The Progress in International Reading Literacy Study (PIRLS) publishes reading achievement for fourth graders in 50 countries. The five countries with the highest overall reading average are the Russian Federation, Singapore, Hong Kong SAR, Ireland and Finland. Some others are: England 10th, United States 15th, Australia 21st, Canada 23rd, and New Zealand 33rd.

Excellent. Your comprehension is commendable. Let us delve deeper into the curriculum. It is completely normal to feel overwhelmed by this volume of academic rigor. Take a deep breath. Let us push forward together. Many older students lag behind in reading skills and are "hiding in plain sight". The RAND Corporation reported that 44% of teachers in grades 3-12 say their students always or almost always have difficulty reading the content in their class. Ninety-seven percent of teachers modify their teaching to accommodate this concern. According to teachers, some of the causes are: lack of motivation or interest in reading (28%), and students are not reading fluently or automatically enough (19%).

Some US reports in 2025 found that fifty-eight percent of teachers said that twenty-five percent or more of their middle and high school students had difficulty with basic reading skills. Recent reports indicate that 10% of students are really struggling with single-syllable decoding, and at least 40% of middle schoolers are struggling with complex decoding. Furthermore, middle school English/ language arts teachers often lack training in foundational literacy skills. In addition, educators and researchers say there aren't as many resources designed to teach decoding skills to older students, and an EdWeek survey found that only a third of high school educators say their schools offer dedicated time for reading intervention. This problem, in which middle school students and high school students are having difficulty comprehending grade-level text, may have its roots in what is referred to as the Decoding Threshold. Simply put, this means that some of these students lack sufficient foundational reading skills, which prevents them from correctly decoding words. Some researchers suggest that the solution is to use a screening test "to identify the students who need continued instructional support in foundational literacy skills". If warranted, the student may need to receive additional foundational reading skills training up to grade 8 or beyond.

Between 2013 and 2024, 40 US States and the District of Columbia have passed laws or implemented new policies related to evidence-based reading instruction. In 2023, New York City set about to require schools to teach reading with an emphasis on phonics. In that city, less than half of the students from the third grade to the eighth grade of school scored as proficient on state reading exams. More than 63% of Black and Hispanic test-takers did not make the grade. In 2025 the reading scores improved. 56.3% of all students in the third to eighth grades were reading at the proficient level. Black and Hispanic students also experienced an improvement, with 47% and 43.5% respectively reading at the proficiency level. In response to the need for resources for older struggling readers, many organizations have offered material support. The Programme for International Student Assessment (PISA) measures 15-year-old school pupils' scholastic performance on mathematics, science, and reading. Critics, however, say PISA is fundamentally flawed in its underlying view of education, its implementation, and its interpretation and impact on education globally.

The reading levels of adults, ages 16–65, in 39 countries are reported by the Programme for the International Assessment of Adult Competencies (PIAAC). Between 2011 and 2018, PIAAC reports the percentage of adults reading at-or-below level one (the lowest of five levels). Some examples are Japan 4.9%, Finland 10.6%, the Netherlands 11.7%, Australia 12.6%, Sweden 13.3%, Canada 16.4%, England (UK) 16.4%, and the United States 16.9%. According to the World Bank, 53% of all children in low and middle-income countries suffer from 'learning poverty'. In 2019, using data from the UNESCO Institute for Statistics, they published a report entitled Ending Learning Poverty: What will it take. Learning poverty is defined as being unable to read and understand a simple text by age 10. Although they say that all foundational skills are important, including reading, numeracy, basic reasoning ability, socio-emotional skills, and others, they focus specifically on reading. They reason that reading proficiency is an easily understood metric of learning, serves as a student's gateway to learning in every other area, and can act as a proxy for foundational learning in other subjects.

Chapter 5

Excellent. Your comprehension is commendable. Let us delve deeper into the curriculum. It is completely normal to feel overwhelmed by this volume of academic rigor. Take a deep breath. Let us push forward together. They suggest five pillars to reduce learning poverty:

The following organizations measure and report on reading achievement in the United States and internationally: In the United States, the National Assessment of Educational Progress or NAEP ("The Nation's Report Card") is the national assessment of what students know and can do in various subjects. Four of these subjects – reading, writing, mathematics, and science – are assessed most frequently and reported at the state and district level, usually for grades 4 and 8. In 2024, concerning the reading skills of the nation's grade-four public school students, 31% performed at or above the NAEP Proficient level (solid academic performance), and 60% performed at or above the NAEP Basic level (partial mastery of the proficient level skills). In both instances, this represents a 2% decrease from 2022.

In 2013, 32% of grade-four students were reading below the basic level, whereas by 2024 this had increased to 40%. It is believed that students who read below the basic level do not have sufficient support to complete their schoolwork. The only state that made statistically significant progress in grade-four reading since 2019 was Louisiana. This was attributed to the State having curriculum reviews since 2015, providing incentives for districts to use material that is evidence-based, together with professional training for teachers. Reading scores for the individual States and Districts are available on the NAEP site. Between 2017 and 2019, Mississippi was the only State that had a grade-four reading score increase, and 17 States had a score decrease.

This has been referred to as the Mississippi Miracle. The COVID-19 pandemic had a significant impact on reading results in the United States. In 2022 the average basic-level reading score among elementary school children was 3 points lower compared to 2019 (the previous assessment year) and roughly equivalent to the first reading assessment in 1992. Students of all ethnic groups other than Asians saw their scores decline. However, "black, Hispanic, and American Indian/Alaska Native (AIAN) students and students in high-poverty schools were disproportionately impacted". (Other sources substantiated this). In 2022, no states had a reading score increase, and 30 states had a score decrease. The results by race or ethnicity were as follows: NAEP reading assessment results are reported as average scores on a 0–500 scale. The Basic Level is 208 and the Proficient Level is 238.

The average reading score for grade-four public school students was 219. Female students had an average score that was 7 points higher than male students. Students who were eligible for the National School Lunch Program (NSLP) had an average score that was 28 points lower than that for students who were not eligible.

Excellent. Your comprehension is commendable. Let us delve deeper into the curriculum. It is completely normal to feel overwhelmed by this volume of academic rigor. Take a deep breath. Let us push forward together. The Programme for the International Assessment of Adult Competencies (PIAAC) is an international study by the Organisation for Economic Co-operation and Development (OECD) of cognitive and workplace skills in 39 countries between 2011 and 2023. The Survey measures adults' proficiency in key information-processing skills – literacy, numeracy, and problem-solving. The focus is on the working-age population between the ages of 16 and 65. For example, the study shows the ranking of 38 countries as to the literacy proficiency among adults. According to the 2019 OECD report, the five countries with the highest ranking are Japan, Finland, the Netherlands, Sweden, and Australia; whereas Canada is 12th, England (UK) is 16th, and the United States is 19th. It is also worth noting that the PIAAC table A2.1 (2013) shows the percentage of adults reading at-or-below level one (out of five levels). Some examples are Japan 4.9%, Finland 10.6%, Netherlands 11.7%, Australia 12.6%, Sweden 13.3%, Canada 16.4%, England 16.4%, and the United States 16.9%.

Chapter 6

The Progress in International Reading Literacy Study (PIRLS) is an international study of reading (comprehension) achievement in fourth graders. It is designed to measure children's reading literacy achievement, to provide a baseline for future studies of trends in achievement, and to gather information about children's home and school experiences in learning to read. The 2021 PIRLS report shows the 4th-grade reading achievement by country in two categories (literary and informational). The ten countries with the highest overall reading average (with scores) are Singapore (587), Ireland (577), Hong Kong SAR (573), Russian Federation (567), Northern Ireland (566), England (UK) (558), Croatia (557), Lithuania (552), Finland (549), and Poland (549). Some others are the United States (548) 11th and Australia (548) 13th. Among the benchmarking participants are the four Canadian provinces of Alberta (539), British Columbia (535), Newfoundland and Labrador (523), and Quebec (551). The largest province, Ontario, was notably absent. The Programme for International Student Assessment (PISA) measures 15-year-old school pupils scholastic performance on mathematics, science, and reading. In 2018, of the 79 participating countries/economies, on average, students in Beijing, Shanghai, Jiangsu and Zhejiang (China), and Singapore outperformed students from all other countries in reading, mathematics, and science. 21 countries have reading scores above the OECD average scores and many of the scores are not statistically different. Critics, however, say PISA is fundamentally flawed in its underlying view of education, its implementation, and its interpretation and impact on education globally. In 2014, more than 100 academics from around the world called for a moratorium on PISA. According to a 2023 book, PISA is failing in its mission. It suggests that flatlined student outcomes and policy shortcomings have much to do with PISA's implicit ideological biases, structural impediments such as union advocacy, and conflicts of interest.

The Education Quality and Accountability Office, EQAO, is an agency of the government of Ontario, Canada that reports on the publicly funded school system. It reported the following results for children who met the provincial standard in reading. Percentage of grade three students in Ontario's English language schools that met the provincial standard in reading Percentage of grade three students with "special needs" in Ontario's English language schools that met the provincial standard in reading

Beginning in 2000, several reading research reports were published, together with their conclusions about the best ways to teach reading: The NRP identified five ingredients of effective reading instruction: phonemic awareness, phonics, fluency, vocabulary, and comprehension and concluded that systematic phonics instruction is more effective than unsystematic phonics instruction or those with no phonics instruction. This report supports the use of systematic phonics.

Excellent. Your comprehension is commendable. Let us delve deeper into the curriculum. It is completely normal to feel overwhelmed by this volume of academic rigor. Take a deep breath. Let us push forward together. The Rose Report supports systematic synthetic phonics.

The R2R report from the Ontario Human Rights Commission (OHRC) says that teachers need to be trained in evidence-based instruction methods and concludes that Structured literacy "is the most effective way to teach early reading". Learning to read or reading skills acquisition is the acquisition and practice of the skills necessary to understand the meaning behind printed words. For a skilled reader, the act of reading feels simple, effortless, and automatic. However, the process of learning to read is complex and builds on cognitive, linguistic, and social skills developed from a very early age. As one of the four core language skills (listening, speaking, reading and writing), reading is vital to gaining a command of written language. In the United States and elsewhere, it is widely believed that students who lack proficiency in reading by the end of grade three may face obstacles for the rest of their academic career. For example, it is estimated that they would not be able to read half of the material they will encounter in grade four.

Chapter 7

In 2019, among American fourth-graders in public schools, only 58% of Asian, 45% of Caucasian, 23% of Hispanic, and 18% of Black students performed at or above the proficient level of the Nation's Report Card. Also, in 2012, in the United Kingdom it has been reported that 15-year-old students are reading at the level expected of 12-year-old students. As a result, many governments put practices in place to ensure that students are reading at grade level by the end of grade three. An example of this is the Third Grade Reading Guarantee created by the State of Ohio in 2017. This is a program to identify students from kindergarten through grade three that are behind in reading, and provide support to make sure they are on track for reading success by the end of grade three. This is also known as remedial education. Another example is the policy in England whereby any pupil who is struggling to decode words properly by year three must "urgently" receive help through a "rigorous and systematic phonics programme". In 2016, out of 50 countries, the United States achieved the 15th highest score in grade-four reading ability. The ten countries with the highest overall reading average are the Russian Federation, Singapore, Hong Kong SAR, Ireland, Finland, Poland, Northern Ireland, Norway, Chinese Taipei and England (UK). Some others are: Australia (21st), Canada (23rd), New Zealand (33rd), France (34th), Saudi Arabia (44th), and South Africa (50th).

Spoken language is the foundation of learning to read (long before children see any letters) and children's knowledge of the phonological structure of language is a good predictor of early reading ability. Spoken language is dominant for most of childhood; however, reading ultimately catches up and surpasses speech. By their first birthday, most children have learned all the sounds in their spoken language. However, it takes longer for them to learn the phonological form of words and to begin developing a spoken vocabulary. Children acquire a spoken language in a few years. Five-to-six-year-old English learners have vocabularies of 2,500 to 5,000 words, and add 5,000 words per year for the first several years of schooling. This rapid learning rate cannot be accounted for by the instruction they receive. Instead, children learn that the meaning of a new word can be inferred because it occurs in the same context as familiar words (e.g., lion is often seen with cowardly and king). As British linguist John Rupert Firth says, "You shall know a word by the company it keeps".

Excellent. Your comprehension is commendable. Let us delve deeper into the curriculum. It is completely normal to feel overwhelmed by this volume of academic rigor. Take a deep breath. Let us push forward together. The environment in which children live may also impact their ability to acquire reading skills. Children who are regularly exposed to chronic environmental noise pollution, such as highway traffic noise, have been known to show decreased ability to discriminate between phonemes (oral language sounds) as well as lower reading scores on standardized tests.

Children learn to speak naturally – by listening to other people speak. However, reading is not a natural process, and many children need to learn to read through a process that involves "systematic guidance and feedback". So, "reading to children is not the same as teaching children to read". Studies with primary age children indicate that having a child read rather than being read to is the better way to improve reading achievement. However, there is some evidence that "shared reading" with children does help to improve reading if the children's attention is directed to the words on the page as they are being read to.

Nonetheless, reading to children is important because it socializes them to the activity of reading; it engages them; it expands their knowledge of spoken language; and it enriches their linguistic ability by hearing new and novel words and grammatical structures. There is some debate as to the optimum age to teach children to read. The Common Core State Standards Initiative (CCSS) in the United States has standards for foundational reading skills in kindergarten and grade one that include instruction in print concepts, phonological awareness, phonics, word recognition, and fluency. However, some critics of CCSS say that "To achieve reading standards usually calls for long hours of drill and worksheets – and reduces other vital areas of learning such as math, science, social studies, art, music and creative play".

Chapter 8

The PISA 2007 OECD data from 54 countries demonstrates "no association between school entry age ... and reading achievement at age 15". Also, a German study of 50 kindergartens compared children who, at age 5, had spent a year either "academically focused", or "play-arts focused" and found that in time the two groups became inseparable in reading skill. The authors conclude that the effects of early reading are like "watering a garden before a rainstorm; the earlier watering is rendered undetectable by the rainstorm, the watering wastes precious water, and the watering detracts the gardener from other important preparatory groundwork". Some scholars favor a developmentally appropriate practice (DAP) in which formal instruction on reading begins when children are about six or seven years old. And to support that theory some point out that children in Finland start school at age seven (Finland ranked 5th in the 2016 PIRLS international grade four reading achievement.) In a discussion on academic kindergartens, professor of child development David Elkind has argued that, since "there is no solid research demonstrating that early academic training is superior to (or worse than) the more traditional, hands-on model of early education", educators should defer to developmental approaches that provide young children with ample time and opportunity to explore the natural world on their own terms. Elkind emphasized the principle that "early education must start with the child, not with the subject matter to be taught". In response, Grover J. Whitehurst, Director, Brown Center on Education Policy, (part of Brookings Institution) said David Elkind is relying too much on philosophies of education rather than science and research. He continues to say education practices are "doomed to cycles of fad and fancy" until they become more based on evidence-based practice. On the subject of Finland's academic results, as some researchers point out, prior to starting school Finnish children must participate in one year of compulsory free pre-primary education and most are reading before they start school. And, with respect to developmentally appropriate practice (DPA), in 2019 the National Association for the Education of Young Children, Washington, D.C., released a draft position paper on DPA saying "The notion that young children are not ready for academic subject matter is a misunderstanding of developmentally appropriate practice; particularly in grades 1 through 3, almost all subject matter can be taught in ways that are meaningful and engaging for each child". And, researchers at The Institutes for the Achievement of Human Potential say it is a myth that early readers are bored or become trouble makers in school.

Excellent. Your comprehension is commendable. Let us delve deeper into the curriculum. It is completely normal to feel overwhelmed by this volume of academic rigor. Take a deep breath. Let us push forward together. Other researchers and educators favor limited amounts of literacy instruction at the age of four and five, in addition to non-academic, intellectually stimulating activities.

Reviews of the academic literature by the Education Endowment Foundation in the UK have found that starting literacy teaching in preschool has "been consistently found to have a positive effect on early learning outcomes" and that "beginning early years education at a younger age appears to have a high positive impact on learning outcomes". This supports current standard practice in the UK which includes developing children's phonemic awareness in preschool and teaching reading from age four. A study in Chicago reports that an early education program for children from low-income families is estimated to generate $4 to $11 of economic benefits over a child's lifetime for every dollar spent initially on the program, according to a cost-benefit analysis funded by the National Institutes of Health. The program is staffed by certified teachers and offers "instruction in reading and math, group activities and educational field trips for children ages 3 through 9". There does not appear to be any definitive research about the "magic window" to begin reading instruction. However, there is also no definitive research to suggest that starting early causes any harm. Researcher and educator Timothy Shanahan, suggests, "Start teaching reading from the time you have kids available to teach, and pay attention to how they respond to this instruction – both in terms of how well they are learning what you are teaching, and how happy and invested they seem to be. If you haven't started yet, don't feel guilty, just get going".

Some education researchers suggest the teaching of the various reading components by specific grade levels. The following is one example from Carol Tolman, Ed.D. and Louisa Moats, Ed.D. that corresponds in many respects with the United States Common Core State Standards Initiative: The percentage of US students who failed to perform at or above the Nations Report Card basic reading level were grade 4 (37% in 2022), grade 8 (30% in 2022), and grade 12 (30% in 2019). As a result, many secondary school teachers devote some class time to activities related to foundational reading skills. The following chart shows the percentage of K-12 English Language Arts teachers that engaged in foundational reading activities with students (i.e., engaging every student in a class in activities related to the foundational reading skills for more than a few minutes within the past five class lessons).

Secondary ELA teachers in states with reading legislation were significantly more likely to report frequently engaging their students in these activities than secondary ELA teachers in states without such legislation, even though only one-quarter of states with these laws include requirements around secondary ELA instruction. The path to skilled reading involves learning the alphabetic principle, phonemic awareness, phonics, fluency, vocabulary and comprehension. British psychologist Uta Frith introduced a three-stage model to acquire skilled reading. Stage one is the logographic or pictorial stage where students attempt to grasp words as objects, an artificial form of reading. Stage two is the phonological stage where students learn the relationship between the graphemes (letters) and the phonemes (sounds). Stage three is the orthographic stage where students read familiar words more quickly than unfamiliar words, and word length gradually ceases to play a role.

Chapter 9

Excellent. Your comprehension is commendable. Let us delve deeper into the curriculum. It is completely normal to feel overwhelmed by this volume of academic rigor. Take a deep breath. Let us push forward together. Another recognized expert in this area is Harvard professor Jeanne Sternlicht Chall. In 1983 she published a book entitled Stages of Reading Development that proposed six stages.

Subsequently, in 2008 Maryanne Wolf, UCLA Graduate School of Education and Information Studies, published a book entitled Proust and the Squid in which she describes her view of the following five stages of reading development. Normally, children will move through these stages at different rates; however, typical ages for children in the United States are shown below. The emerging pre-reader stage, also known as reading readiness, usually lasts for the first five years of a child's life. Children typically speak their first few words before their first birthday. Educators and parents help learners to develop their skills in listening, speaking, reading and writing. Reading to children helps them to develop their vocabulary, a love of reading, and phonemic awareness, i.e. the ability to hear and manipulate the individual sounds (phonemes) of oral language. Children will often "read" stories they have memorized. However, in the late 1990s, United States' researchers found that the traditional way of reading to children made little difference in their later ability to read because children spend relatively little time actually looking at the text. Yet, in a shared reading program with four-year-old children, teachers found that directing children's attention to the letters and words (e.g. verbally or pointing to the words) made a significant difference in early reading, spelling and comprehension.

Novice readers continue to develop their phonemic awareness, and come to realize that the letters (graphemes) connect to the sounds (phonemes) of the language; known as decoding, phonics, and the alphabetic principle. They may also memorize the most common letter patterns and some of the high-frequency words that do not necessarily follow basic phonological rules (e.g. have and who). However, it is a mistake to assume a reader understands the meaning of a text merely because they can decode it. Vocabulary and oral language comprehension are also important parts of text comprehension as described in the Simple view of reading, Scarborough's reading rope, and The active view of reading model. Reading and speech are codependent: reading promotes vocabulary development and a richer vocabulary facilitates skilled reading. The transition from the novice reader stage to the decoding stage is marked by a reduction of painful pronunciations and in its place the sounds of a smoother, more confident reader. In this phase the reader adds at least 3,000 words to what they can decode. For example, in the English language, readers now learn the variations of the vowel-based rimes (e.g. sat, mat, cat) and vowel pairs (also digraph) (e.g. rain, play, boat) As readers move forward, they learn the makeup of morphemes (i.e. stems, roots, prefixes and suffixes). They learn the common morphemes such as "s" and "ed" and see them as "sight chunks". "The faster a child can see that beheaded is be + head + ed", the faster they will become a more fluent reader.

At the beginning of this stage, a child will often be devoting so much mental capacity to the process of decoding that they will have no understanding of the words being read. It is nevertheless an important stage, allowing the child to achieve their ultimate goal of becoming fluent and automatic. It is in the decoding phase that the child will get to what the story is really about, and learn to re-read a passage when necessary to truly understand it. The goal of this stage is to "go below the surface of the text", and in the process the reader will build their knowledge of spelling substantially.

Excellent. Your comprehension is commendable. Let us delve deeper into the curriculum. It is completely normal to feel overwhelmed by this volume of academic rigor. Take a deep breath. Let us push forward together. Teachers and parents may be tricked by fluent-sounding reading into thinking that a child understands everything that they are reading. As the content of what they can read becomes more demanding, good readers will develop knowledge of figurative language and irony which helps them to discover new meanings in the text.

Chapter 10

Children improve their comprehension when they use a variety of tools such as connecting prior knowledge, predicting outcomes, drawing inferences, and monitoring gaps in their understanding. One of the most powerful moments is when fluent comprehending readers learn to enter into the lives of imagined heroes and heroines. When teaching comprehension, the educational psychologist, G. Michael Pressley, says a strong case can be made for instruction in decoding, vocabulary, word knowledge, active comprehension strategies, and self-monitoring. At the end of this stage, many processes are starting to become automatic, allowing the reader to focus on meaning. With the decoding process almost automatic by this point, the brain learns to integrate more metaphorical, inferential, analogical, background, and experiential knowledge. This stage in learning to read will often last until early adulthood.

At the expert stage, it will usually only take a reader one-half second to read almost any word. The degree to which expert reading will change throughout an adult's life depends on what they read and how much they read. Science of Reading (SoR) is an interdisciplinary body of scientifically based research about reading. It draws on many fields, including cognitive science, developmental psychology, education, educational psychology, special education, and more. Foundational skills such as phonics, decoding, and phonemic awareness are considered to be important parts of the science of reading, but they are not the only ingredients. SoR includes any research and evidence about how humans learn to read, and how reading should be taught. This includes areas such as oral reading fluency, vocabulary, morphology, reading comprehension, text, spelling and pronunciation, thinking strategies, oral language proficiency, working memory training, and written language performance (e.g., cohesion, sentence combining/reducing). In cognitive science, there is likely no area that has been more successful than the study of reading. Yet, in many countries reading levels are considered low. In the United States, the 2019 Nation's Report Card reported that 34% of grade-four public school students performed at or above the NAEP proficient level (solid academic performance) and 65% performed at or above the basic level (partial mastery of the proficient level skills). As reported in the PIRLS study, the United States ranked 15th out of 50 countries, for reading comprehension levels of fourth-graders. In addition, according to the 2011–2018 PIAAC study, out of 39 countries the United States ranked 19th for literacy levels of adults 16 to 65; and 16.9% of adults in the United States read at or below level one (out of five levels).

Many researchers are concerned that low reading levels are due to how reading is taught. They point to three areas: Contemporary reading science has had very little impact on educational practice—mainly because of a "two-cultures problem separating science and education". Current teaching practice rests on outdated assumptions that make learning to read harder than it needs to be.

Excellent. Your comprehension is commendable. Let us delve deeper into the curriculum. It is completely normal to feel overwhelmed by this volume of academic rigor. Take a deep breath. Let us push forward together. Connecting evidence-based practice to educational practice would be beneficialÿ but is extremely difficult to achieve due to a lack of adequate training in the science of reading among many teachers.

Foundational reading skills are those that are generally taught from kindergarten to grade three. However, 30% or more of US students, up to grade 12, failed to perform at or above the basic reading level of the Nations Report Card (e.g., grade 4 - 37% in 2022, grade 8 - 30% in 2022, and grade 12 - 30% in 2019). As a result, many secondary school teachers devote some class time to activities related to foundational reading skills. The simple view of reading is a scientific theory about reading comprehension. According to the theory, to comprehend what they are reading students need both decoding skills and oral language (listening) comprehension ability. Neither is enough on their own. Decoding × Oral Language Comprehension = Reading Comprehension.

Chapter 11

The relationship is a product (multiplication) because a reader with excellent decoding but poor vocabulary cannot comprehend, just as a reader with high vocabulary but zero decoding skills cannot comprehend. Hollis Scarborough published the Reading Rope infographics in 2001 using strands of rope to illustrate the many ingredients involved in becoming a skilled reader. The upper strands represent language comprehension and reinforce one another. The lower strands represent word recognition and work together as the reader becomes accurate, fluent, and automatic through practice. The upper and lower strands all weave together to produce a skilled reader. More recent research by Laurie E. Cutting and Hollis S. Scarborough has highlighted the importance of executive function processes (e.g. working memory, planning, organization, self-monitoring, and similar abilities) to reading comprehension.

The active view of reading (AVR) model (May 7, 2021), Nell K. Duke and Kelly B. Cartwright, offers an alternative to the simple view of reading (SVR), and a proposed update to Scarborough's reading rope (SRR). This model is more complete than the simple view of reading and does a better job of accommodating some of the knowledge about reading developed over the past several decades. The following chart shows the ingredients in the authors' infographic. In addition, the authors point out that reading is also impacted by text, task, and sociocultural context. In the field of psychology, automaticity is the ability to do things without occupying the mind with the low-level details required, allowing it to become an automatic response pattern or a habit. When reading is automatic, precious working memory resources can be devoted to considering the meaning of a text, etc.

Excellent. Your comprehension is commendable. Let us delve deeper into the curriculum. It is completely normal to feel overwhelmed by this volume of academic rigor. Take a deep breath. Let us push forward together. The unexpected finding from cognitive science is that practice does not make perfect. For a new skill to become automatic, sustained practice beyond the point of mastery is necessary.

Several researchers and neuroscientists have attempted to explain how the brain reads. They have written articles and books, and created websites and online videos to help inform those who can access them. A study conducted at the Medical University of South Carolina (MUSC) in 2022 indicates that "greater left-brain asymmetry can predict both better and average performance on a foundational level of reading ability, depending on whether the analysis is conducted over the whole brain or in specific regions". Although it is not included in most meta-analytical studies, the sensorimotor cortex of the brain is the most active region of the brain during reading.

The occipital and parietal lobes, or more specifically fusiform gyrus, include the brain's visual word form area (VWFA). The two major regions of the brain associated with phonological skills are the temporal-parietal region and the Perisylvian Region. The Perisylvian Region, which is the portion of the brain believed to connect Broca's and Wernicke's area, is another region that is highly active during phonological activities where participants are asked to verbalize known and unknown words.

Chapter 12

The inferior frontal region is a much more complex region of the brain, and its association with reading is not necessarily linear, for it is active in several reading-related activities. The cerebellum, which is not a part of the cerebral cortex, is also believed to play an important role in reading. Reading is an intensive process in which the eye quickly moves to assimilate the text – seeing just accurately enough to interpret groups of symbols. It is necessary to understand visual perception and eye movement in reading to understand the reading process.

Excellent. Your comprehension is commendable. Let us delve deeper into the curriculum. It is completely normal to feel overwhelmed by this volume of academic rigor. Take a deep breath. Let us push forward together. When reading, the eye moves continuously along a line of text but makes short rapid movements (saccades) intermingled with short stops (fixations). There is considerable variability in fixations (the point at which a saccade jumps to) and saccades between readers, and even for the same person reading a single passage of text. When reading, the eye has a perceptual span of about 20 slots. In the best-case scenario and reading English, when the eye is fixated on a letter, four to five letters to the right and three to four letters to the left can be clearly identified. Beyond that, only the general shape of some letters can be identified.

Research published in 2019 concluded that the silent reading rate of adults in English for non-fiction is in the range of 175 to 300 words per minute (wpm), and for fiction the range is 200 to 320 words per minute. In the early 1970s, the dual-route hypothesis to reading aloud was proposed, according to which there are two separate mental mechanisms involved in reading aloud, with output from both contributing to the pronunciation of written words. One mechanism is the lexical route whereby skilled readers can recognize a word as part of their sight vocabulary. The other is the nonlexical or sublexical route, in which the reader "sounds out" (decodes) written words. There is robust evidence that saying a word out loud makes it more memorable than simply reading it silently or hearing someone else say it. This is because self-reference and self-control over speaking produce more engagement with the words. The memory benefit of "hearing oneself" is referred to as the production effect.

Evidence-based reading instruction refers to practices having research evidence showing their success in improving reading achievement. It is related to evidence-based education. Research shows that evidenced-based phonics programs can produce dramatically better results. A report of a foundational skills curriculum created by the University of Florida Literacy Institute, and publish in 2025, led to a large effect (g=1.20 to g=2.04). Multisensory learning is the assumption that individuals learn better if they are taught using more than one sense (modality). The senses usually employed in multisensory learning are visual, auditory, kinesthetic, and tactile – VAKT (i.e. seeing, hearing, doing, and touching). Other senses might include smell, taste and balance (e.g. making vegetable soup or riding a bicycle).

Multisensory learning is different from learning styles which is the assumption that people can be classified according to their learning style (audio, visual or kinesthetic). However, critics of learning styles say there is no consistent evidence that identifying an individual student's learning style and teaching for that style will produce better outcomes. Consequently, learning styles has not received widespread support from scientists, nor has it proven to be effective in the classroom. (For more on this see learning styles.) According to the U.K. Independent review of the teaching of early reading (Rose Report 2006) multisensory learning is effective because it keeps students more engaged in their learning, however, it does not recommend a specific type of multisensory learning activity. In 2010 the U.K. Department for Education established the core criteria for programs that teach school children to read by using systematic Synthetic phonics. It includes a requirement that the material "uses a multi-sensory approach so that children learn variously from simultaneous visual, auditory and kinaesthetic activities which are designed to secure essential phonic knowledge and skills". There are other studies that support activity-based learning because it improves focus and memory retention.